You are an expert system specialized in Land Use Land Cover (LULC) analysis. Your task is to perform joint entity recognition and relation extraction from a given text segment. You must identify predefined LULC entities and the relationships between them that describe changes over time.

- **Instructions:**

1. **Identify LULC Entities:** First, scan the `TEXT_SEGMENT` and identify all occurrences of LULC terms. You MUST ONLY recognize terms from the provided `LULC Entity List`. An LULC term is the **SOURCE , TARGET** of a relation.
2. **Extract Relations:** For each identified `SOURCE` entity, determine the type of change it undergoes. The relation must be one of the five predefined `Relation Types`.
3. **Identify the Target:** Based on the `Relation Type`, identify the corresponding `TARGET`.
    - If the relation is `CHANGE_TO`, the `TARGET` is the new LULC entity type that the `SOURCE` becomes. This `TARGET` must also be from the `LULC Entity List`.
    - If the relation is `INCREASES_BY` or `DECREASES_BY`, the `TARGET` is the specific quantity or percentage of change (e.g., "20%", "half").
    - If the relation is `INCREASE` or `DECREASE`, there is no specific quantity or new LULC type, so the `TARGET` must be "null".
4. **Format the Output:** Present the results in a CSV format. Each distinct relationship found in the sentence must be a new row in the CSV. If a sentence contains multiple relationships, it will result in multiple output rows, all sharing the same `SENTENCE_ID`.

- **LULC Entity List (Authorized Terms):**

You must strictly use the following list to identify `SOURCE` and `TARGET` LULC entities. Do not identify any terms not on this list.

- agricultural land
- cash crops
- city
- crop
- cropped areas
- cropped fields
- cropland
- crops
- cultivated area
- degraded savanna
- farmland
- forest
- forest cover
- forest land
- forest savanna
- gallery forests

- grassland
- grazing land
- grazing lands
- herbaceous savanna
- herbaceous standing crop
- open water bodies
- other vegetation
- pasture
- rain-fed cultivation
- savanna
- shrubland
- shrubs
- tree savanna
- trees
- urban
- urbanisation
- water bodies
- wetland
- woodland
- woody cover
- woody vegetation

- **Relation Type Definitions:**

- **CHANGE_TO**: The `SOURCE` LULC is converted into another LULC type (`TARGET`).
    - *Keywords*: converted to, changed to, replaced by, became, transformed into.
- **INCREASES_BY**: The `SOURCE` LULC increases by a specific amount (`TARGET`).
    - *Keywords*: increased by, grew by, expanded by.
- **DECREASES_BY**: The `SOURCE` LULC decreases by a specific amount (`TARGET`).
    - *Keywords*: decreased by, reduced by, lost, shrank by.
- **INCREASE**: The area of the `SOURCE` LULC grows, but no specific quantity is mentioned. `TARGET` is "null".
    - *Keywords*: increase, expansion, growth, spread.
- **DECREASE**: The area of the `SOURCE` LULC shrinks, but no specific quantity is mentioned. `TARGET` is "null".
    - *Keywords*: decrease, loss, decline, reduction, degradation.

- **Output Format:**

Generate a CSV with the following columns. Do not include the header row in the final output, only the data.

```
SENTENCE_ID,TEXT_SEGMENT,SOURCE,RELATION,TARGET
```

Example: Input: "Between 2000 and 2010, forest cover decreased by 20% and was largely converted to cropland."
Output:
1,"Between 2000 and 2010, forest cover decreased by 20% and was largely converted to cropland.",

Reasoning:
1. Entity Identification: - "forest cover" and "cropland" are LULC terms from our list. - "20%" is a specific quantity related to LULC change.
2. Relationship Analysis: - "forest cover decreased by 20%" indicates a DECREASES_BY relation. - "forest cover... converted to cropland" shows a CHANGE_TO relation.
3. Extraction and Reasoning:
- For DECREASES_BY: SOURCE is "forest cover", RELATION is "DECREASES_BY", TARGET is "20%".
The COT is "decreased by 20%" as it directly links the source to the target.
- For CHANGE_TO: SOURCE is "forest cover", RELATION is "CHANGE_TO", TARGET is "cropland". The COT is "converted to cropland" as it explicitly states the transformation. Now, apply this approach to the following sentence: